

# Final Internship Project Submission

## BCG Data Science Project

### Project Name

BCG Data Science Project – Customer Churn Analysis and Prediction

### Project Description

This project focuses on analysing customer churn and building a predictive machine-learning model to identify customers who are likely to leave a business. The work follows an end-to-end data science workflow, including exploratory data analysis (EDA), data cleaning, feature engineering, model training, and model evaluation.

### Approach

- Explored the customer dataset and identified important patterns and relationships.
- Performed data preparation and cleaning to create a usable modelling dataset.
- Created engineered features to improve the predictive information available to the model.
- Split the prepared dataset into training and testing data using a 75:25 split.
- Trained a Random Forest Classifier for customer churn prediction.
- Evaluated the model using Accuracy, Precision, Recall, and F1 Score.

### Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and Jupyter Notebook.

### Model Evaluation

| Metric    | Result |
|-----------|--------|
| Accuracy  | 0.9039 |
| Precision | 0.8261 |
| Recall    | 0.0519 |
| F1 Score  | 0.0977 |

Note: The evaluation values above are taken from the completed modelling notebook.

### GitHub Link

[https://github.com/Mahanand4/BCG\\_Data\\_Science\\_Project](https://github.com/Mahanand4/BCG_Data_Science_Project)

**Submission Note:** The project files, datasets used for modelling, notebooks, and project README have been uploaded to the GitHub repository.